Mini co-pilot work split and milestones

We're building the two-person mini co-pilot option for COSC 329. This plan gives each person a starting area and keeps the shared experiment work visible. The split is proposed; agree on it together before treating it as a commitment.

Roles

David - data and n-grams: collect and document Python projects, prepare the dataset, implement word/character models and lead repeatable experiment runs. Start with T04 and T06.

Kamso - syntax and editor: investigate incomplete-code parsing, build the AST baseline and editor, and lead syntax integration in the hybrid. Start with T08 and T13 using a fake predictor.

Both - evaluation and delivery: agree on prediction inputs and scored targets, define metrics, build the hybrid, compare two models, review each other's work and prepare the report/demo. Shared work is split into a named lead before coding.

Milestones

Dates are in 2026, Pacific time. Internal targets are our proposed schedule, not extra instructor deadlines. Canvas due dates below come from the course assignment list; check the corresponding assignment page for format requirements.

M1 Repository ready

Team target: Oct 30. Repository link due Nov 1 at 11:59 p.m.

Done when: Repository and planning files are ready; agree on roles; submit an accessible repository URL.

M2 Dataset and first models

Team target: Nov 8. Internal target.

Done when: 10+ sources logged; stable project splits; character and word models work.

M3 Structure and experiments

Team target: Nov 15. Internal target.

Done when: AST baseline and five-fold runner work; failures and metrics recorded.

M4 Hybrid and editor

Team target: Nov 22. Internal target.

Done when: Hybrid evaluated; editor shows five suggestions and can switch models.

M5 Presentation ready

Team target: Nov 27. Presentation due Nov 29 at 11:59 p.m.

Done when: Compare two models; record demo; check the separate Canvas presentation instructions.

M6 Final handoff

Team target: Dec 11. Final deliverables due Dec 13 at 11:59 p.m.

Done when: Reproduce results; finish report; check and submit the required final files.

Settle these first

- Confirm Python as both the dataset language and implementation language.

- Agree on what counts as a word target and how partial-word completion works.

- Confirm how the general 70/30 split relates to the two-person five-fold requirement. Our proposal is a 30% project-level final holdout and five development folds within the remaining 70%; it needs instructor confirmation. Ten total projects would leave very small folds, so consider a larger dataset.

- Use only training-fold vocabulary and statistics when fitting a model. Every model receives the same prefix and hidden target boundary. Never give a predictor the rest of the file.

Work independently

Before implementing models, agree on a shared prediction interface. One possible shape is fit(source_strings) and suggest(prefix, limit=5), but this is a proposal, not existing code. David can work on tiny sample files while Kamso uses a fake predictor to build the editor. Document shared changes in INTERFACES.md before using them on both branches.

What done means

Relevant tests pass, another partner has reviewed the work, and setup instructions match the code. Dataset choices and experiment settings are recorded. Results come from saved runs; example output is not evidence of model quality.

Keep one active task per person where possible. Use short updates: finished, next, blocked. The task IDs and acceptance checks live in BACKLOG.md.

Scope

We need character and word n-grams for n=1..10, an AST model, a hybrid and a simple editor with five suggestions. Keep the editor basic. Chat, cloud model services, accounts and a full IDE extension are outside this plan.

Course guidance: https://cmps-people.ok.ubc.ca/bowenhui/analytics/project.html

Project backlog

This is our working task list, not a list of extra course requirements. All tasks below are still Todo. No models, tests or demo code have been implemented in this starter.

Workflow: Todo -> In progress -> Review -> Done. Pick one task at a time, make a branch named after its ID (for example t06-word-model), and link the pull request to the task. Move it to Done after the other person reviews it and its completion checks pass.

David owns the data and n-gram track, Kamso owns the parser and editor track, and Both means one leads the implementation while the other reviews it. Agree on the lead before starting shared tasks.

Epics and user stories

- Setup: As teammates, we want a shared repo and agreed interfaces so we can work separately and combine our work.

- Data: As experimenters, we want documented, repeatable source data so comparisons use the same inputs.

- Models: As a coder, I want useful next-word suggestions so I spend less time typing repeated code.

- Evaluation: As experimenters, we want fair measurements so we can explain which model works best.

- Editor: As a coder, I want five selectable suggestions while typing so I can try the models in practice.

- Delivery: As a team, we want reproducible results and a clear demo so others can understand our work.

Tasks

T01 Set up the shared repository

- Status: Todo

- Epic: Setup | Owner: David | Milestone: M1 | Priority: High

- Depends on: None

- Done when: Both partners can access the repo; planning files are committed; both understand the first tasks.

T02 Agree on roles and prediction format

- Status: Todo

- Epic: Setup | Owner: Both | Milestone: M1 | Priority: High

- Depends on: None

- Done when: Confirm Python, ownership and full-word suggestion format; record decisions in INTERFACES.md.

T03 Clarify evaluation split

- Status: Todo

- Epic: Setup | Owner: David | Milestone: M1 | Priority: High

- Depends on: None

- Done when: Ask how 70/30 and five-fold CV fit together; record instructor guidance before formal experiments.

T04 Choose and log dataset sources

- Status: Todo

- Epic: Data | Owner: David | Milestone: M2 | Priority: High

- Depends on: T02

- Done when: Log at least 10 independent Python projects with URL, license, exact commit and included paths.

T05 Prepare repeatable dataset and splits

- Status: Todo

- Epic: Data | Owner: David | Milestone: M2 | Priority: High

- Depends on: T03 T04

- Done when: Exclude generated/vendor code consistently; deduplicate; save project-level splits, seed and file hashes; keep raw code out of Git.

T06 Build word n-gram models

- Status: Todo

- Epic: Models | Owner: David | Milestone: M2 | Priority: High

- Depends on: T02

- Done when: Support n=1..10, smoothing and unseen contexts; document tokenization; test on small known sequences.

T07 Build character n-gram models

- Status: Todo

- Epic: Models | Owner: David | Milestone: M2 | Priority: High

- Depends on: T02

- Done when: Support n=1..10; generate full-word candidates with length/stop limits; expose the agreed suggestion interface.

T08 Try a parser on incomplete code

- Status: Todo

- Epic: Models | Owner: Kamso | Milestone: M2 | Priority: High

- Depends on: T02

- Done when: Compare candidate parser behaviour on unfinished expressions; select an error-tolerant approach and document failure cases.

T09 Build the AST prediction baseline

- Status: Todo

- Epic: Models | Owner: Kamso | Milestone: M3 | Priority: High

- Depends on: T08

- Done when: Produce ranked word candidates from prefix structure; document candidate rules, syntax checks and fallback; never use hidden future text.

T10 Build shared contexts and metrics

- Status: Todo

- Epic: Evaluation | Owner: Both | Milestone: M3 | Priority: High

- Depends on: T02

- Done when: Score the same held-out word targets for every model; verify exact-match accuracy and character edit distance with hand-worked cases.

T11 Run five-fold model comparisons

- Status: Todo

- Epic: Evaluation | Owner: David | Milestone: M3 | Priority: High

- Depends on: T05 T06 T07 T10

- Done when: Fit on each training fold only; save per-fold results for every n plus mean and sample SD; record runtime settings and failures.

T12 Evaluate AST and hybrid models

- Status: Todo

- Epic: Evaluation | Owner: Both | Milestone: M4 | Priority: High

- Depends on: T09 T10 T11

- Done when: Kamso leads syntax integration; David leads experiment runs; compare AST and a documented hybrid with n-gram baselines using the same contexts.

T13 Build a basic editor with fake predictions

- Status: Todo

- Epic: Editor | Owner: Kamso | Milestone: M3 | Priority: High

- Depends on: T02

- Done when: Typing shows up to five suggestions; selecting one replaces the partial word; use a fake model first so work can start immediately.

T14 Connect two models and handle failures

- Status: Todo

- Epic: Editor | Owner: Kamso | Milestone: M4 | Priority: High

- Depends on: T09 T12 T13

- Done when: Switch between two selected models; update suggestions as code changes; handle empty output/parser errors and check typing responsiveness.

T15 Choose models and compare coding experience

- Status: Todo

- Epic: Evaluation | Owner: Both | Milestone: M4 | Priority: High

- Depends on: T11 T12 T14

- Done when: Explain the choice using results; both try the same coding tasks with two models; record useful and bad suggestions and actual observations.

T16 Prepare presentation and demonstration

- Status: Todo

- Epic: Delivery | Owner: Both | Milestone: M5 | Priority: High

- Depends on: T15

- Done when: Show problem, methods, results and limitations; rehearse the demo; confirm duration/format on Canvas before recording.

T17 Finish report and reproduce results

- Status: Todo

- Epic: Delivery | Owner: Both | Milestone: M6 | Priority: High

- Depends on: T15

- Done when: Explain data, splits, methods, hybrid and metrics; include real tables and limitations; another partner reproduces the key run from README.

T18 Review final submission and contributions

- Status: Todo

- Epic: Delivery | Owner: Both | Milestone: M6 | Priority: High

- Depends on: T16 T17

- Done when: Check current Canvas deliverables; confirm teaching-team access; record actual contributions; one submits and the other verifies.

Decisions before we start coding

Nothing in this file is implemented yet. These are questions and proposed boundaries for David and Kamso to settle together.

Predictions

- What should count as a word: identifiers, keywords, numbers, punctuation or some subset?

- How do we represent code before the cursor and a partly typed word?

- What should every model return so the editor can show the same five-suggestion list?

- One possible interface is fit(source_strings) and suggest(prefix, limit=5). Decide before using it across both branches.

Fair comparison

Use the same held-out word boundaries for every model. Predictors receive only the code before the cursor; the hidden next word and later source belong in the evaluator only. Agree on what to do when a parser fails or a model returns no suggestion.

Data and results

Record source URLs, exact versions, licenses and exclusions in the dataset log. Agree on project-level data splits and remove duplicates across splits. Build vocabulary and fit counts on training data only. Save scores per fold before calculating averages and standard deviations.

Suggested file ownership

David: future dataset and character/word model files.

Kamso: future parser, AST model and editor files.

Shared: evaluation, hybrid, dependency changes and report. Agree on a lead for each shared task before coding.

These files will be created when we start the matching backlog tasks. The editor can use temporary fixed suggestions while the models are being built, so neither person has to wait for the other.